ENT-164 · INTRO TO MAKING

Course Syllabus

What this course covers: learning outcomes, the weekly schedule, grading, policies, and student resources.

Fall 2026 | 3 credits | Thursdays 1:20–4:20pm | JCC 301 / Nolop

This syllabus is subject to change. Students will be notified of any changes via Canvas.

Overview

How can we harness maker tools and technology to build working prototypes that address real-world challenges?

This entrepreneurial course demystifies technology by teaching you to build smart device prototypes using physical making, basic electronics, and programming with AI agents. By creating working prototypes, you'll master the iterative innovation process essential to entrepreneurship: **build, test, learn, and improve**. This practical approach transforms technology from an intimidating black box into an accessible tool for bringing your ideas to life. You'll develop an entrepreneurial mindset by learning to work as a team, fail fast, and iterate rapidly.

This class teaches you to use an AI coding agent to program devices and connect them to mobile apps. All without writing a single line of code by hand. No programming experience is required: you'll direct the AI, test on real hardware, and iterate.

> **No restrictions on enrollment.** All undergraduate and graduate students are welcome, including first-year undergrads. No prior experience in design thinking, innovation, or programming is needed.

Instructor & TAs



Thomas Van de Velde

INSTRUCTOR

Thomas is a technology executive and educator with 20+ years leading digital transformations. At Accenture he built digital engineering teams for retailers like Best Buy; at Amazon he led global AI-driven operations and co-founded Amazon Fresh stores and Amazon Locker. Today he is the founder of TeachPilot and lectures in Entrepreneurship & AI Innovation at Tufts, focused on democratizing engineering education through AI-powered, project-based learning.

thomas.van_de_velde@tufts.edu · Office hours: go.tufts.edu/meet_tvdv · More about Thomas →

Alexander Beattie

TA

Mechanical Engineering · Class of 2027
alexander.beattie@tufts.edu

Office hours: by appointment

Patrice Pinardo

TA

Computer Science · Class of 2027
patrice.pinardo@tufts.edu

Office hours: by appointment

Learning outcomes

By the end of this course, you will be able to:

Make & fabricate

- Design and fabricate physical objects using CAD, laser cutters, and 3D printers
- Build functional circuits and program microcontrollers using MicroPython and AI

Connect & integrate

- Create smart IoT devices that sense, visualize, and respond to environmental data
- Integrate multiple technologies into a cohesive working prototype

Collaborate & iterate

- Work effectively in multidisciplinary teams to design and build prototypes
- Apply the "build-measure-learn" loop through rapid iteration and failing fast

Communicate & present

- Present working prototypes to peers and public audiences
- Articulate design decisions and lessons learned

How class works

This course is intended to be taught in person. We expect students to make every effort to join us in the (physical) classroom unless they encounter extenuating circumstances. Important maker, prototyping, and programming skills will be demonstrated in class. If you need to miss class, you will need to make an appointment with an instructor or a TA to help you catch up.

Each class is run as a workshop. We meet in the classroom for the first 60–90 minutes to recap the previous class, introduce foundational maker skills, and complete short exercises. We then move to Nolop for the rest of the session, where the work is hands-on: applying those skills and building with your team.

Milestone reflections

> **Reflection is a critical part of the learning process.** You'll write five reflections, one at each milestone that closes out a phase of the course: after 3D printing (Week 4), after connectivity (Week 6), after the Robo Challenge (Week 8), at the mid-point of the final project (Week 11), and a final reflection on the whole course (Week 13). Each one is a short pause to examine what you learned, what challenged you, and how you'll apply it. Submit yours by the end of the day it is due, at go.tufts.edu/reflect. Reflections are part of your participation grade.

Course modules

The course is divided into two modules.

Module 1 · Foundational skills

Learn to make physical things using hand making techniques, CAD tools, 3D printers, and laser cutting. You'll work on basic electronics projects to enable your projects to interact with the world. In Week 2, you'll be placed into teams of 4–5 students and collaborate on the **Robo Challenge**: design and build a robot that can move as many balls as possible between two lines. You'll also start defining the final project you'll present on demo day.

Module 2 · Final project work

Build on the first module's foundation to develop your team's final project. You'll learn advanced skills including data sensing, visualization, and AI integration. Your team chooses the real-world challenge you want to tackle. Your final project must demonstrate the core maker techniques: laser cutting, 3D printing, electronics, connectivity, and AI. The module ends with a public demo day.

Course schedule

Thirteen weeks, from kickoff to the final showcase. Dates are Fall 2026.

WEEK 1

Thu, Sep 10

Introductions and Course Overview

Welcome to the course. Meet your classmates and instructors, explore the makerspace, and get your first taste of hands-on making.

- Identify key equipment and safety protocols in the makerspace
- Complete a simple hands-on build from start to finish

Workshop 1 · Introductions and Course Overview

WEEK 2

Thu, Sep 17

Hand Making and Robot Challenge Kickoff

Learn foundational techniques for turning ideas into tangible prototypes using sketching and physical mockups.

- Understand the Robo Challenge requirements and form your team
- Rapidly sketch 3+ concept variations for your design
- Build a foam core mockup to test form and mechanics

Workshop 2 · Hand Making and Robot Challenge Kickoff

👤 · Onboarding survey. Share your background and goals

WEEK 3

Thu, Sep 24

Laser Cutting

Transition from hand-making to digital fabrication by designing in CAD and cutting with a laser.

- Take measurements of your foam core prototype and the parts it will hold
- Create a laser-ready 2D design file from a sketch
- Laser cut structural components

Workshop 3 · Laser Cutting

WEEK 4

Thu, Oct 1

3D Printing

Expand your digital fabrication skills to three dimensions with CAD modeling and 3D printing.

- Model 3D parts (mounts, brackets, custom components)
- 3D print functional parts for your project

Workshop 4 · 3D Printing

👤 · Laser cut project. Present a functional laser-cut creation

WEEK 5

Thu, Oct 8

Electronics

Bring your projects to life with electronics and learn to program microcontrollers using AI-assisted coding.

- Wire motors, sensors, and controls on a breadboard
- Program basic movements using MicroPython and AI

Workshop 5 · Electronics

👤 · 3D print project. Present a functional 3D-printed part

👥 · Assembly checkpoint. Check in with your team on the robot

WEEK 6

Thu, Oct 15

Going Wireless

Make your devices communicate wirelessly and enable remote monitoring and control.

- Connect a microcontroller to a wireless network
- Build a remote control interface

Workshop 6 · Going Wireless

👤 · Electronics project. Present a working circuit

👤 · Physical making reflection

WEEK 7

Thu, Oct 22

Final Robot Assembly and Testing

Apply everything you've learned to build a working robot with your team.

- Integrate all components into a fully functional robot
- Debug and iterate based on testing results

Workshop 7 · Final Robot Assembly and Testing

👤 · Digital Making reflection

WEEK 8

Thu, Oct 29

Robo Challenge and Team Retrospective

Put your robot to the test in a competitive race and reflect on your team's journey.

- Demonstrate a working robot under real-world conditions
- Identify what worked, what didn't, and what you'd do differently

Workshop 8 · Robo Challenge and Team Retrospective

👥 · Robo Challenge. Compete and demonstrate your team's robot

👤 · Milestone reflection. After the Robo Challenge

WEEK 9

Thu, Nov 5

The Final Project Sprint

A three-hour sprint: from idea to a prototype plan. Teams brainstorm, choose a concept, and leave with the plan they'll build for Demo Day.

- Generate and validate a product concept with your team
- Pitch your idea and incorporate peer feedback

Workshop 9 · The Final Project Sprint

👤 · Peer review. Feedback on classmates' ideas

👤 · Peer evaluations. Rate your teammates' contributions

👤 · Self review. Reflect on your Robo Challenge contributions

WEEK 10

Thu, Nov 12

From Device to Cloud

Learn to gather data from the physical world and present it in meaningful visual formats.

- Collect environmental data using sensors
- Display sensor data in a real-time visualization
- Apply sensing techniques to your final project

Workshop 10 · From Device to Cloud

👥 · Project checkpoint 1. Core functionality and technical approach

👥 · Product concept. Pitch your team's idea to the class

WEEK 11

Thu, Nov 19

Final Project Design Review

Add intelligence to your hardware projects by integrating AI capabilities.

- Integrate AI-powered decision making into a physical device
- Identify opportunities to enhance your final project with AI

Workshop 11 · Final Project Design Review

👥 · Project checkpoint 2. Integrated features and user experience

👤 · Intelligent device project. Present an AI-powered device

👤 · Milestone reflection. Final project mid-point

BREAK

Thu, Nov 26

Thanksgiving. No class

WEEK 12

Thu, Dec 3

Final Project Making Time

Dedicated workshop time to complete and polish your final project before the showcase.

- Complete a functional prototype ready for demonstration
- Prepare a compelling presentation of your product

Workshop 12 · Final Project Making Time

WEEK 13

Thu, Dec 10

Final Project Showcase

Celebrate your accomplishments by presenting your finished product to the Tufts community.

- Present and demonstrate your finished product to an audience
- Reflect on your growth and lessons learned throughout the course

Workshop 13 · Final Project Showcase

👤 · Peer evaluation. Assess teammates' contributions

👥 · Final product demo. Present a working prototype

👤 · Self review. Reflect on your growth

👤 · Milestone reflection. The course, end to end

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Grading

How your grade is weighted, and the policies around assignments.

Individual assignments	**35%**
Team assignments, presentations, and reports *(team grades are shared by default but may be adjusted based on individual performance)*	**35%**
Peer and self evaluations (mid-term and final)	**20%**
Reflections and participation	**10%**

> **Peer and self evaluations.** Twice during the semester (mid-term and final), you'll assess your own contributions and those of your teammates. These evaluations help ensure fair grading for team projects and encourage accountability within your team. Your feedback is confidential and will be used to adjust individual grades when contributions are unequal.

Assignment policies

- ✓ All assignments are submitted through **Canvas**.
- ✓ Assignments are typically due on **Tuesdays by noon** the day of class.
- ✓ There will be a **25% late fee** for submissions received during the grace period (within 1 week of the deadline). We will not accept submissions beyond the grace period.
- ✓ **Missed work due to illness:** if you are sick and unable to complete an assignment on time, contact the instructor before the deadline to discuss an extension.
- ✓ **Missed presentations:** if you are too ill to present in class, we can arrange for you to present remotely or reschedule.

Course materials

All materials are provided by the course at no cost to students. You will receive an electronics starter kit at the beginning of the semester. Additional materials and tools will be available at Nolop. No textbooks are required.

What to bring to class

- Laptop (required)
- USB-C adapter if your laptop only has USB-A ports
- External mouse (recommended for CAD work)
- Your electronics kit (bring to every class)

Online platforms and tools

- **Canvas** for assignment submissions, course materials, announcements, and updates
- **Claude / Gemini / ChatGPT** for code generation and electronics
- **Onshape** for 2D and 3D modelling
- **opencode** for MicroPython programming. Writing code and running it on your devices

Norms & policies

Attendance policy

- ✓ On-time, in-person class attendance and respectful class participation is required.
- ✓ If you need to miss class, arrive late, leave early (e.g., to attend a job interview), or attend remotely due to extenuating circumstances (e.g., illness, travel overseas for family reasons), notify instructors by **5:30pm the day before class**. You may also use the Short-Term Illness Notification tool in SIS.
- ✓ **Absences due to illness:** if you are too ill to attend class, inform the instructor as soon as possible. You do not need to provide medical documentation. Please do not come to class if you are sick; we will work with you to catch up on missed material.
- ✓ When doing group work, observe team process norms including equal participation of all team members. These norms will be introduced at the beginning of the semester.
- ✓ Students are expected to adhere to the school's honor code.
- ✓ Diversity is celebrated at Tufts, as it should be around the world. Diversity is multidimensional and spans gender identity, sexual orientation, race, prior experiences, and more. Respectful, inclusive, and equitable treatment of all students and instructors is a fundamental norm in this class.

Academic integrity

Tufts holds its students strictly accountable for adherence to academic integrity. The consequences for violations can be severe. It is critical that you understand the requirements of ethical behavior and academic work as described in Tufts' Academic Misconduct policy. If you ever have a question about the expectations concerning a particular assignment or project in this course, be sure to ask one of the instructors for clarification.

While collaboration is encouraged, individual assignments are expected to be submitted individually using the components provided to the student. Allocate sufficient time to request replacement components if required. It is your responsibility to pick up new components promptly when missing a class. Submitting an individual assignment as shared work will result in points being divided among the participants.

On the use of AI

This course integrates AI as a core tool in the maker's workflow. You will learn to use AI to design, iterate, and build more advanced prototypes than would otherwise be possible. **AI use is encouraged and expected.** You may use tools like Claude, ChatGPT, Gemini, and other AI assistants for:

- Generating and debugging code for microcontrollers

- Brainstorming design ideas
- Understanding technical concepts
- Troubleshooting electronics and fabrication issues

Attribution is not required, but documenting your AI-assisted workflow is encouraged as a learning practice.

Student resources

Support is available. Please use it.

Religious accommodations

Tufts University faculty, staff, and administration highly value and acknowledge the religious diversity of its student body. Students seeking religious accommodations related to their holy days are encouraged to collaborate with faculty to make arrangements during the first week of each semester. See the Multifaith Calendar for upcoming holidays, the University Religious Accommodations Policy, and members of the University Chaplaincy, who are available to respond to questions on religious observances.

Accommodations for disabilities

Tufts is committed to providing equal access and support to all qualified students through the provision of reasonable accommodations. If you have a disability that requires reasonable accommodations, contact the StAAR Center at StaarCenter@tufts.edu or 617-627-4539. Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect of their provision.

Academic support at StAAR

The StAAR Center offers **free** resources to all students: writing appointments, subject tutoring in a variety of disciplines, and academic coaching for skills like time management and navigating procrastination. You may make an appointment for any of these services by visiting students.tufts.edu/staar-center.

Mental health & well-being

As a student, there may be times when personal stressors or difficulties interfere with your academic performance or well-being. The Dean of Students Office offers support and care to undergraduates and graduate students who are experiencing difficulties, and can also aid faculty in their work with students. Through Tufts' Counseling and Mental Health Service (CMHS) you can access mental health support 24/7; CMHS also provides information on additional resources, confidential consultation, brief counseling, and urgent care at no cost for all Tufts undergraduates, and for graduate students who have paid the student health fee. Call 617-627-3360 or visit go.tufts.edu/Counseling.

Ears for Peers

Tufts' anonymous and confidential student-run hotline, open every night 7pm–7am during the academic year. Call or text 617-627-3888.

Dean of Students

Personal stressors or difficulties interfering with your academic performance or well-being? The Dean of Students Office can help. Get in touch.